GROUP PROJECT 2

Project Group: 9284
Black-Litterman Model

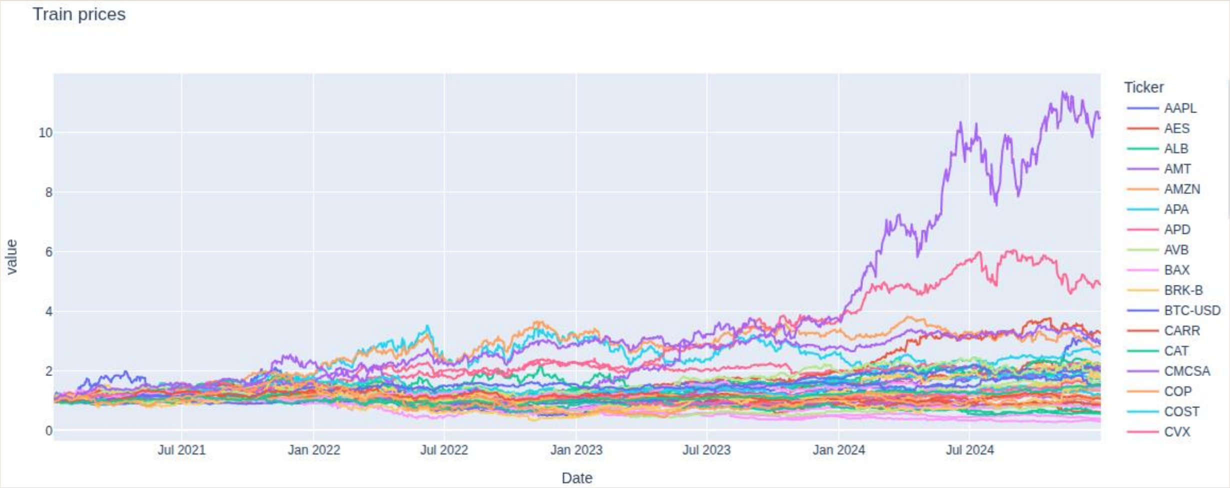
Step: Data & Preprocessing

- Training Period: 2021-01-04 to 2024-12-31
- Testing Period: 2025-01-06 to 2025-05-08
- Data: S&P 500 stocks, Gold, Bitcoin
- Return formula: r_t = log(P_t / P_{t−1})
- Used DGS2 series from FRED as daily risk-free rate

Tickers/Companies

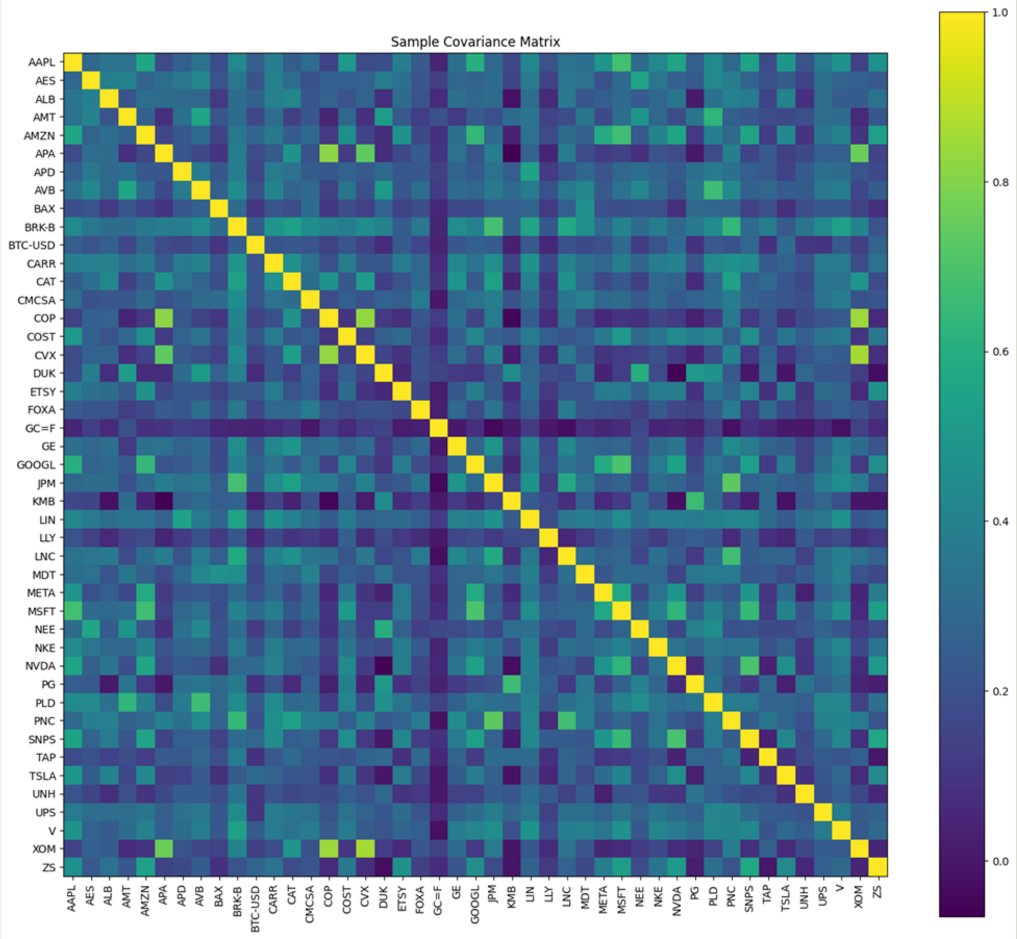
Sectors/company					
Materials	LIN (Linde plc)	APD (Air Products and Chemicals Inc.)	ALB (Albemarle Corporation)		
Communication Services	META (Meta Platforms Inc.)	CMCSA (Comcast Corporation)	FOXA (Fox Corporation)	GOOGL (Alphabet Inc.)	
Energy	XOM (Exxon Mobil)	COP (ConocoPhillips)	APA (APA Corporation)	CVX (Chevron Corporation)	
Financials	JPM (JPMorgan Chase & Co)	PNC (PNC Financial Services)	LNC (Lincoln National)	BRK-B (Berkshire Hathaway)	V (Visa Inc.)
Industrials	CAT (Caterpillar Inc.)	UPS (United Parcel Service)	CAPR (Carrier Global Corporation)	GE (General Electric)	
Information Technology	NVDA (NVIDIA	SNPS (Synopsys Inc.)	ZS (Zscaler Inc.)	AAPL (Apple Inc.)	MSFT (Microsoft)

Train & Test Returns:



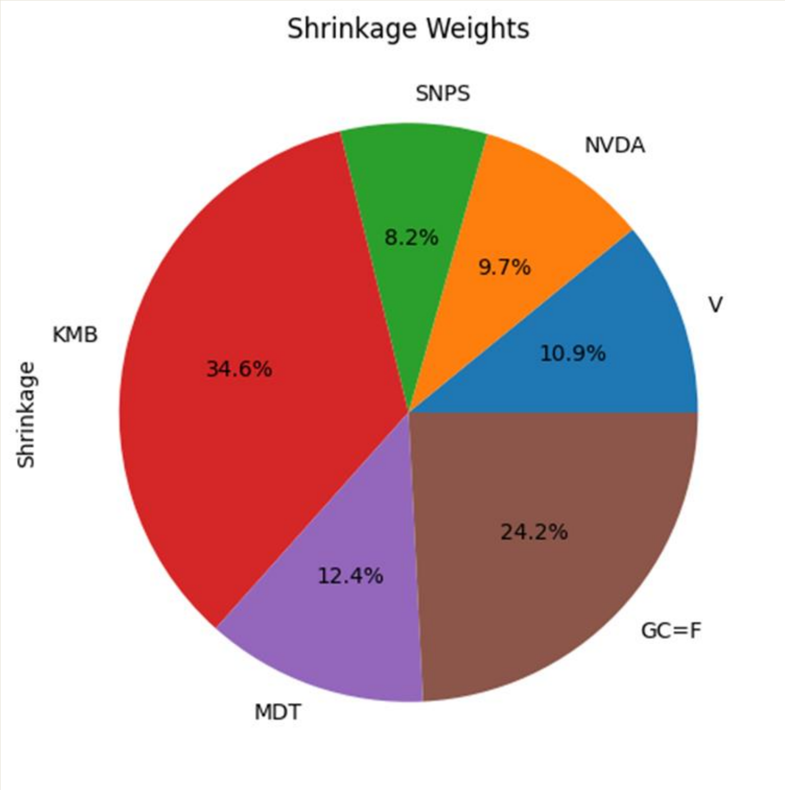
Covariance Estimators: 2 ways

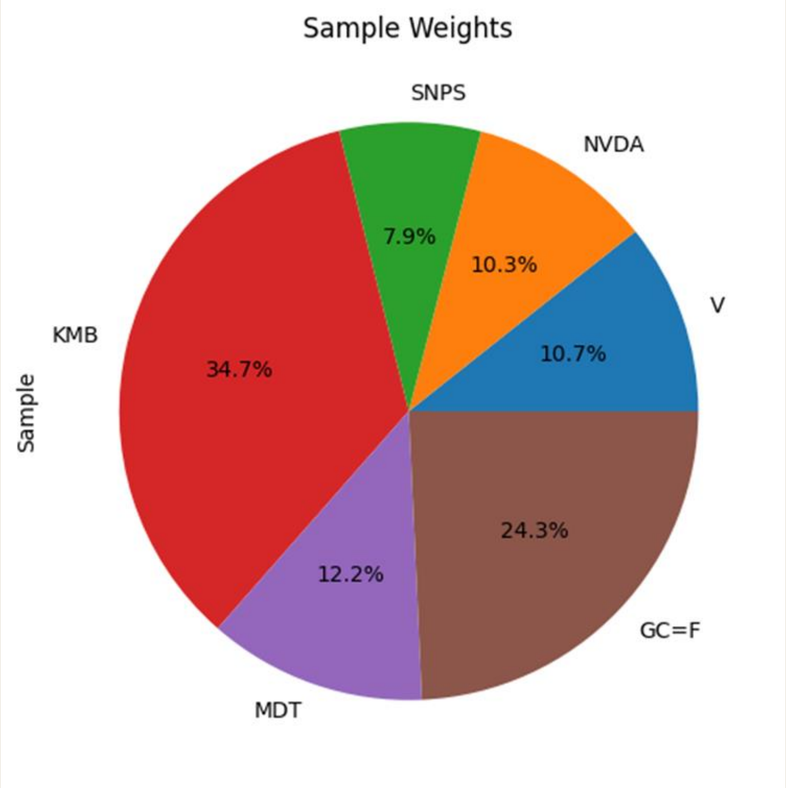
- Sample estimator: Σ_sample (variance and covariance)
- Ledoit–Wolf shrinkage estimator: Σ_shrinkage



Step: Markowitz Optimization

- Mean-variance optimization (MVO) used
- Constraints: long-only, max weight 0.35
- Objective: maximize Sharpe ratio
- Implemented using SLSQP from scipy.optimize





MVO: Portfolio Performance

Portfolio stats:

	SampleΣ		ShrinkΣ	
	Train	Test	Train	Test
μ_ann	0.074	0.181	0.071	0.185
σ_ann	0.130	0.176	0.129	0.175
Sharpe	0.572	1.029	0.553	1.059
TotalReturn	0.469	0.070	0.451	0.072

Risk contribution table (train Σ):

Asset	SampleΣ	ShrinkΣ
GC=F	9.0%	9.5%
KMB	27.6%	28.7%
MDT	10.8%	11.1%
NVDA	27.5%	25.1%
SNPS	13.8%	14.2%
V	11.3%	11.5%

•The Sharpe ratio in the training process is less than one. It usually is interpreted that the portfolio composition is not good and the excess risk is not compensated by enough return.

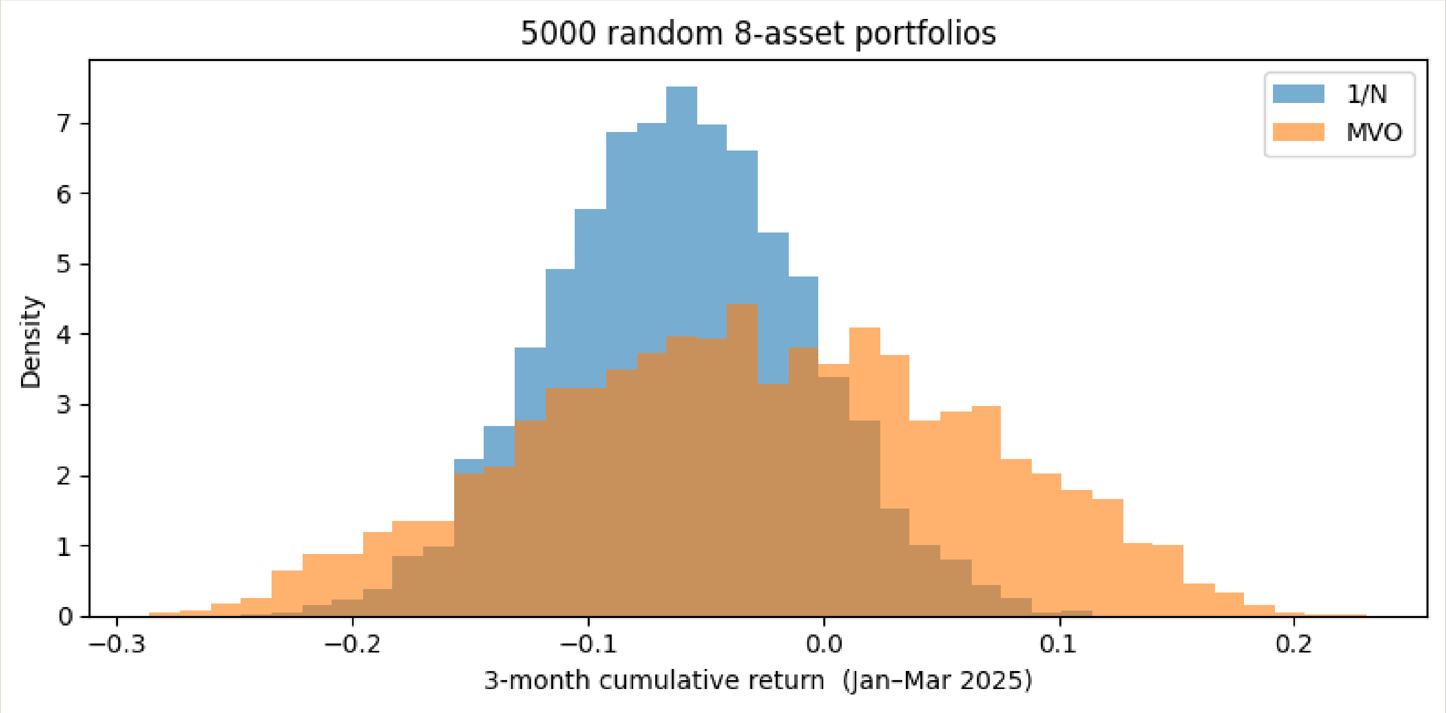
•The volatility in the testing period is substantially higher than the training period. It shows that the market is passing through an unstable phase.

•The total return and also Sharpe ratio are negative for the evaluation (test) period. So, we are losing money!

•Looking at the news we can see why we are facing this situation. In our interpretation, the global financial markets, w.r.to the US, are experiencing a high volatility time due to new tariff policies of the US government.

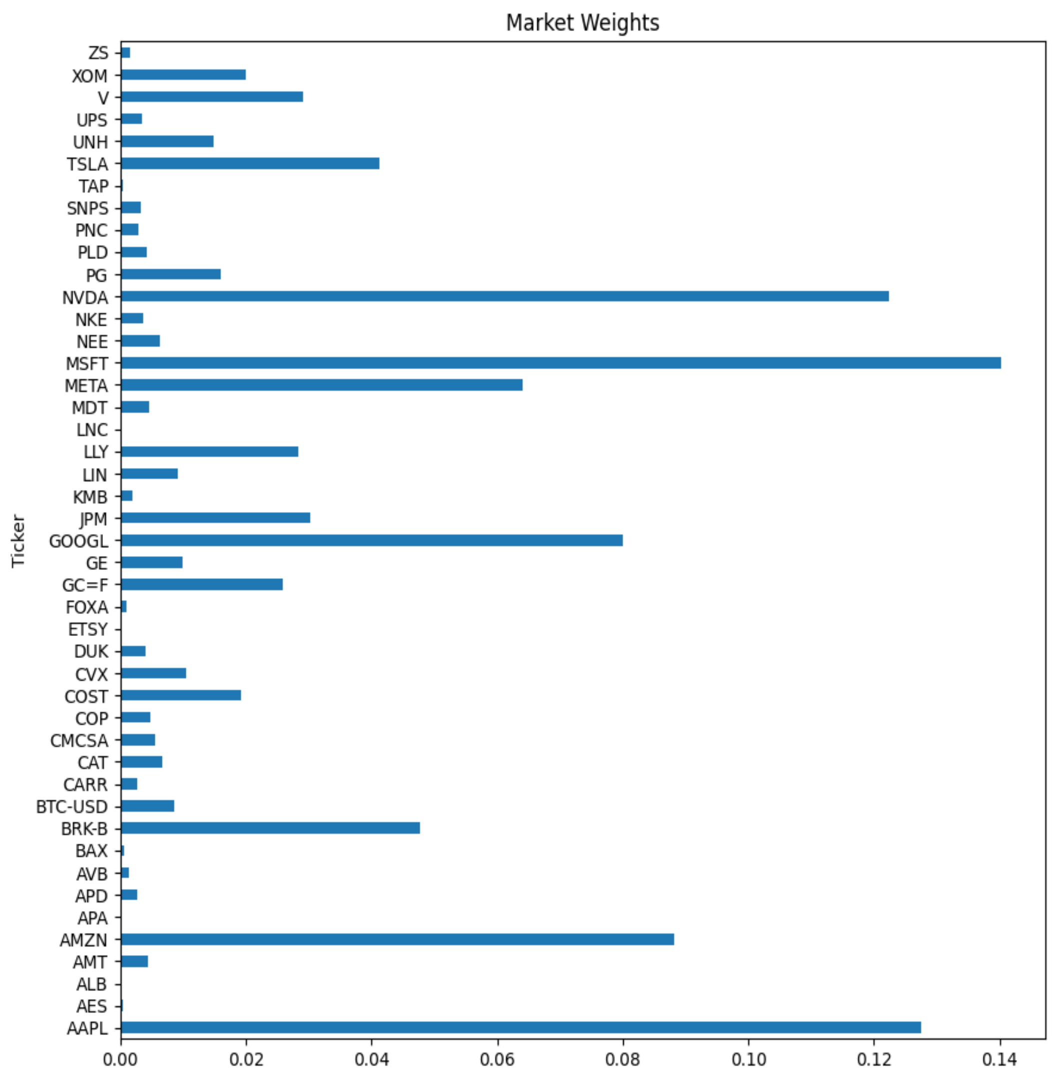
Step: Random Portfolio Strategy

- Randomly choose 8 assets from universe
- Compare equal-weight (1/N) vs MVO optimized weights
- Simulated 5000 portfolios over the test period
- Observation: MVO outperformed 1/N



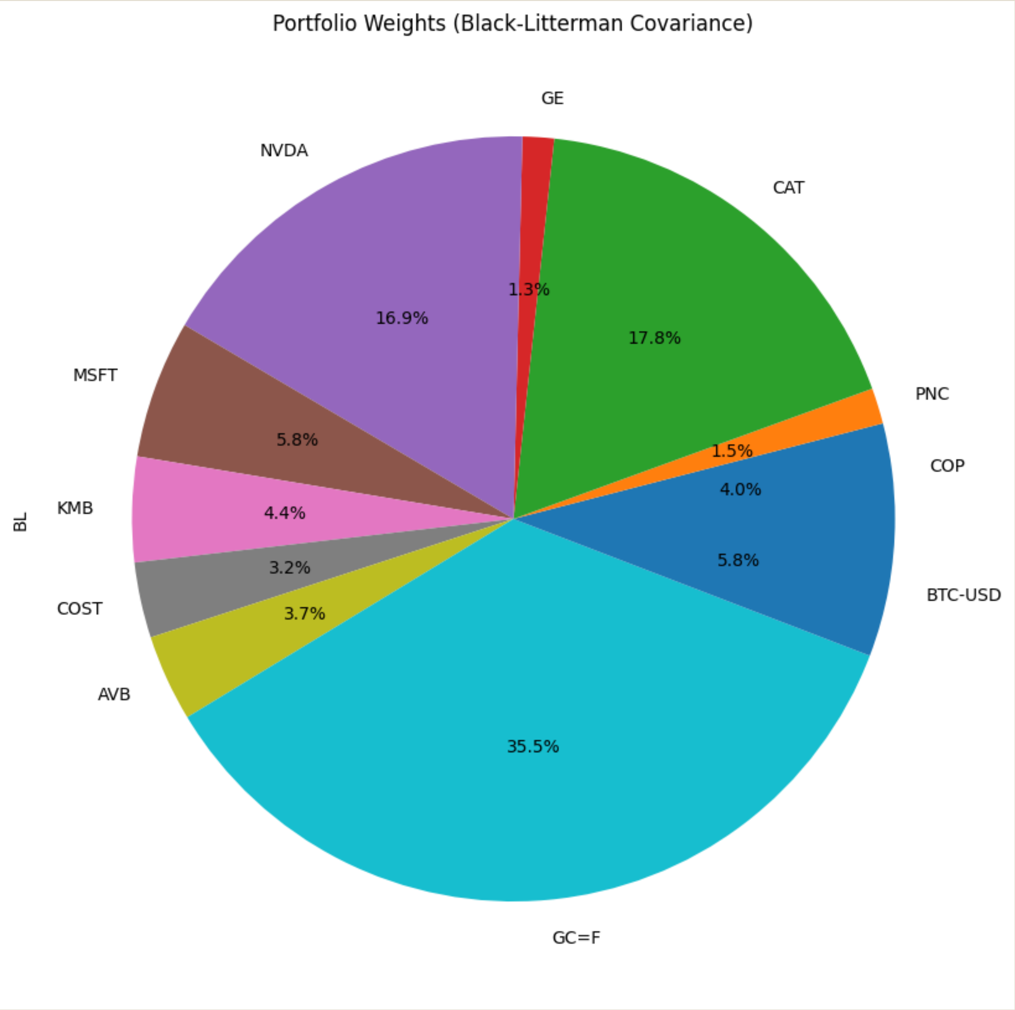
Step: Black-Litterman Model

- Used news-based absolute & relative views
- Defined matrices: P (views), Q (return), Ω (confidence)
- Confidence levels: high = 0.1, med = 0.3, low = 1.0
- Used CAPM-implied returns: $\pi = \delta \Sigma w\_m$



BL Posterior and Optimization

- Posterior returns: μ_BL and covariance: Σ_BL
- Formula: μ_BL = [(τΣ)^−1 + P^TΩ^−1P]^{-1}[...]
- Optimization: maximize Sharpe using μ_BL, Σ_BL
- Pruned weights < 1% for clarity and feasibility



BL Weights Comparison

	MVO_Σsample	MVO_Σshrink	BL
COP	0.0	0.0	0.0402
PNC	0.0	0.0	0.0151
V	0.107	0.109	0.0
CAT	0.0	0.0	0.1776
GE	0.0	0.0	0.0133
NVDA	0.103	0.097	0.1691
SNPS	0.079	0.082	0.0
MSFT	0.0	0.0	0.0584
KMB	0.347	0.346	0.0444
COST	0.0	0.0	0.0319
MDT	0.122	0.124	0.0
AVB	0.0	0.0	0.0366
GC=F	0.243	0.242	0.3551
BTC-USD	0.0	0.0	0.0582

Performance Comparison

- The performance of our portfolio is not as expected.
- Since the beginning of the year 2025 due to the tariff policies of the US government, the financial market has been very volatile and unstable.
- We are experiencing a regime shift in the market. So, the importance of the historical data on the future performance of the portfolio is reduced.

=== In-sample (2021-24) ===

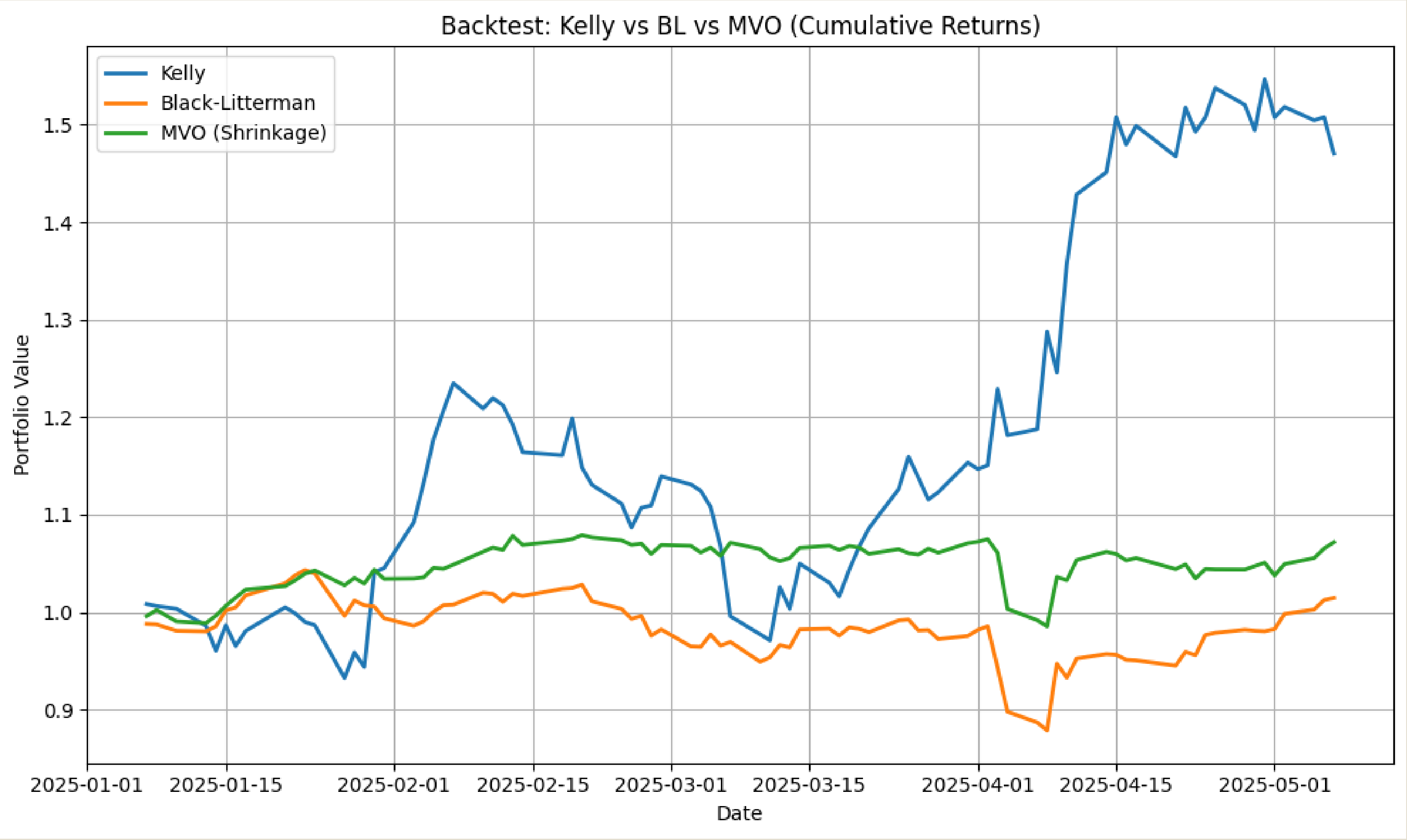
	MarkovΣshrink	BL
μ_ann	0.071	0.185
σ_ann	0.129	0.169
Sharpe	0.553	1.094

=== Out-of-sample (Jan–Mar 2025) =

	MarkovΣshrink	BL
μ_ann	0.185	0.033
σ_ann	0.175	0.242
Sharpe	1.059	0.135

Kelly Criterion: Performance

- Kelly's Sharpe ratio and cumulative return growth better than MVO and BL.



Final Notes

- BL: combines market belief and investor views
- Compared performance to MVO and Kelly under test conditions.
- BL has given max weights to GC=F followed by CAT, NVDA,BTC-USD, MSFT, KMB whereas MVO has given max weights to KMB, followed by GC=F, NVDA,V and MDT.
- Highest contribution to variance for BL model is by NVDA, followed by CAT, BTC-USD and GC=F.
- Risk-return tradeoff: Well, it is said for higher returns one need to take higher risks..
- Highest contribution to variance for MVO is by KMB, NVDA, SNPS, V.MDT and GC=F
- In long run Kelly outperformed MVO and BL both because of many assets going short and ability to fractional Kelly and monthly rebalancing.
- High volatility do impact the prediction ability

Portfolio Optimization: BL vs MVO vs Kelly

- Thank you